

JONATHAN TROWBRIDGE

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CAREER SUMMARY

Principal Software Engineer & Security Researcher with 10+ years of experience building secure, scalable web, mobile, and backend systems. **Certified in OSCP+, GPEN, GCIH, GSEC, and GFACT** with a deep technical foundation in offensive security, original vulnerability research resulting in 6 recent disclosures to Debian package maintainers, and a portfolio focused on re-engineering functional PoCs for known, critical-rated vulnerabilities.

CYBERSECURITY CERTIFICATIONS

OSCP+ Offensive Security Certified Professional+, OffSec	2025
GPEN GIAC Penetration Tester, SANS Technology Institute	2025
GCIH GIAC Certified Incident Handler, SANS Technology Institute	2025
GSEC GIAC Security Essentials, SANS Technology Institute	2025
GFACT GIAC Foundational Cybersecurity Technologies, SANS Technology Institute	2024

VULNERABILITY RESEARCH & DISCLOSURE

- **sohps (Shared Object Hijack Path Scanner)**: Developed a security auditing tool to identify shared object hijack vulnerabilities in Linux ELF binaries and Appliance containers. Used it to scan ~70,000 Debian packages, resulting in 6 disclosed vulnerabilities: Debian Bugs [#1138498 \(python3-torch\)](#), [#1138589 \(libarrow500\)](#), [#1138592 \(gpu-basis-universal\)](#), [#1138596 \(python3-ifcopenshell\)](#), [#1138597 \(python3-ledger\)](#), and [#1138072 \(clang-tools-21\)](#)

EXPLOIT REPRODUCTION

- **CVE-2026-21858 (ni8mare)**: Developed a functional PoC for an unauthenticated RCE in n8n automation platforms (< 1.121.0), demonstrating full system compromise via logic flaws.
- **CVE-2025-55182 (React2Shell)**: Reproduced a critical RCE via vulnerable React Server Components (RSC) behavior, bridging the gap between front-end architecture and server-side exploitation.
- **CVE-2026-33017 (Langflow)**: Engineered a reproducible RCE exploit targeting Langflow AI orchestration layers, highlighting vulnerabilities in emerging LLM infrastructure.
- **CVE-2026-1357 (WPVivid)**: Demonstrated RCE within the WPVivid Backup & Migration WordPress plugin, targeting widespread deployment surfaces.

TECHNICAL SKILLS

Security Research & Engineering: Penetration Testing, Vulnerability Research, Exploit Development, Web & Mobile Application Security, Network Security, API Security, Threat Modeling, Security-Focused Code Review

Languages: Python, Bash, PowerShell, Kotlin, Java, C#, TypeScript, Dart, Go

Frameworks: Flutter, KMP, Xamarin, Ktor, ASP.NET Core, Node.js, React, Angular

Infrastructure & Environments: Linux, Windows Server, Docker, Git, CI/CD Pipeline Security

Databases: SQL Server, MariaDB, Redis, PostgreSQL, SQLite, MongoDB

PROFESSIONAL EXPERIENCE

Lead Principal Software Engineer | Coinme | Seattle, WA (Remote) Sept. 2021 - Oct. 2024

- Directed the native platform migration to Flutter, architecting authentication and payment flows to ensure PCI data security and transaction integrity.
- Engineered a secure mobile architecture using Riverpod, implementing state-management patterns to prevent PII leakage across modular, decoupled feature sets.
- Established a secure SDLC and mentored team members to accelerate the delivery of high-impact features while ensuring architectural integrity and scalability.

Senior Full Stack Software Engineer | CoStar Group | Richmond, VA (Hybrid) Jan. 2019 - Aug. 2021

- Led development of an Android listing module and supporting ASP.NET Core API, ensuring secure data handling and endpoint protection for high-traffic commercial services.
- Engineered scalable web applications with React and Node.js and built an enterprise AWS push notification system for secure, real-time cross-platform communication.
- Partnered cross-functionally to implement secure design principles and code review standards, ensuring resilient and consistent feature releases across mobile and web.

Full Stack Software Engineer | Mapleton Hill Media | Boulder, Colorado Aug. 2016 – Feb. 2018

- Developed cross-platform mobile applications for logistics and agriculture, ensuring secure client-side data handling and architectural consistency.
- Modernized legacy ASP.NET architectures by migrating monolithic APIs to modular services, improving service isolation and system auditability.
- Engineered real-time synchronization for distributed inventory systems, implementing secure data flows and validation logic across mobile devices and APIs.

Full Stack Application Developer | Worldview Solutions Inc. | Richmond, VA Feb. 2013 – Feb. 2016

- Led implementation of claims-based authentication and IAM for a U.S. Air Force recruiting portal, ensuring strict access control and federal compliance.
- Developed regulatory and monitoring modules for Virginia state agencies, focusing on data integrity and workflow efficiency within ASP.NET environments.
- Modernized legacy web architectures to improve performance and maintainability, ensuring architectural consistency across client projects.

EDUCATION

Undergraduate Certificate in Applied Cybersecurity | SANS Technology Institute 2025

Bachelor of Science in Computer Science | Virginia Commonwealth University 2013